ML and NMF for DIABETES DETECTION on the PIMA DATASET

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1 INTRODUCTION

Diabetes is a group of chronic and non-communicable metabolic diseases that affect how the body uses blood sugar. It is characterized by either a high blood sugar level due to insufficient insulin production in the body or ineffective use of insulin. In this project, some machine learning (ML) techniques are used to analyze PIDD (Pima Indian Diabetes Dataset) for the sake of identifying the risk factors of diabetes mellitus.

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2 DATASET DESCRIPTION and MORE

We have used the Pima Indian Diabetes Dataset (PIDD). The data is collected from 768 female patients of Pima Indian heritage and they are 21 years old or older. The dataset is arranged in 9 features which are summarized in Table 1:

Table 1: Features of PIDD and their meaning.

Feature	Meaning
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic BP (mm Hg)
SkinThickness	Triceps skin-fold thickness
Insulin	2-hour serum insulin
BMI	Body mass index
DiabetesPedigreeFunction	Genetic pre-disposition
Age	Age in years
Outcome	Diabetes status (1: diabetic, 0: non-diabetic)

Moreover, we found out that 35% of the patients are diabetic while the remaining are non-diabetic, thereby signaling the existence of a moderate imbalance in the data. The specific patterns of the underlying dataset are presented in Table 2.

Table 2: PIDD specific patterns.

Feature	Diabetics	Non-Diabetics	Insight
Glucose	Much higher	Lower	Strongest early signal
BMI	Higher	Lower	Obesity risk
Age	Higher	Lower	Risk increases with age
Pregnancies	Higher	Lower	Gestational risk
Pedigree	Higher	Lower	Genetic component

3 DATA INSPECTION

At a first glance, it seems that there are no NaNs which could be misleading. There are also a lot of zeros (see Table 3) in the data which makes us a bit confused, as several medical features cannot be zero physiologically. Since we are dealing with PIDD, we need to watch out for the invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI. As suspected, we found a very important information which we summarize in the table below. This table tells us that, among other things, nearly 30% of SkinThickness and 49% of Insulin go missing. Properly treating these incidents will dramatically improve model reliability where as ignoring it will definitely lead to false predictions. Data imputation is needed here to properly deal with the missing values. Since median is robust to skewed medical variables and a best practice in this domain, all the missing values are replaced by the median of their respected features.

Table 3: Invalid zeros in PIDD.

Feature	Number of invalid zeros
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

4 INTERPRETING RISK FACTORS with LR and RF

The `LR` model provided the risk factors in descending order as depicted in Figure 1:

```
Glucose                    1.183444
BMI                        0.709709
Pregnancies                0.373019
DiabetesPedigreeFunction   0.287745
Age                        0.186393
Insulin                   -0.044651
BloodPressure             -0.014548
SkinThickness              0.013877
dtype: float64
```

Figure 1: Risk factors PIDD.

The results depicted in Figure 1 tell the following story:

- Glucose $-->$ strongest predictor
- BMI + Age $-->$ compounding risk
- DiabetesPedigreeFunction $-->$ genetic susceptibility
- Coefficients explain direction and strength of risk

Next, we change the model to the non-linear `RandomForestClassifier()` just to improve performance, see Figure 2:

```
[[85 15]
 [24 30]]
              precision    recall  f1-score   support

           0       0.78      0.85      0.81       100
           1       0.67      0.56      0.61        54

    accuracy                           0.75       154
   macro avg       0.72      0.70      0.71       154
weighted avg       0.74      0.75      0.74       154

ROC-AUC: 0.8160185185185184
```

Figure 2: Calculations with `RandomForestClassifier()` for PIDD.

The results in Figure 2 tell us that RF:

- Captures nonlinear interactions (e.g., Age × BMI)
- Often improves recall slightly
- Less interpretable but stronger predictive power

We have the following key early detection conclusions:

- Glucose is the single most powerful early indicator
- BMI and Age amplify risk together
- Genetic predisposition is non-negligible
- Recall-optimized models reduce missed diabetes cases significantly

5 FEATURE DISTRIBUTION

For PIDD, the most informative "relative plots" are feature distributions split by Outcome (0 vs 1). These let us visually compare diabetic vs non-diabetic patients feature by feature, which is exactly what we want for early detection insight. Since KDE is ideal for medical risk comparison, we have plotted KDE distributions for each feature, separated by Outcome as shown in Figure 3.

Figure 3 depicts the relative density distribution of the features by outcome. The interpretation is done by looking at the rightward shifts and separation of the curves; orange (diabetic) and blue (non-diabetic):

- **Glucose:** Most important feature, diabetic curve shifts strongly right and minimal overlap at high glucose. These imply glucose is the best early detection marker.
- **BMI:** Diabetics concentrated at higher BMI, clear obesity-related risk. This means, BMI is an important modifiable risk factor.

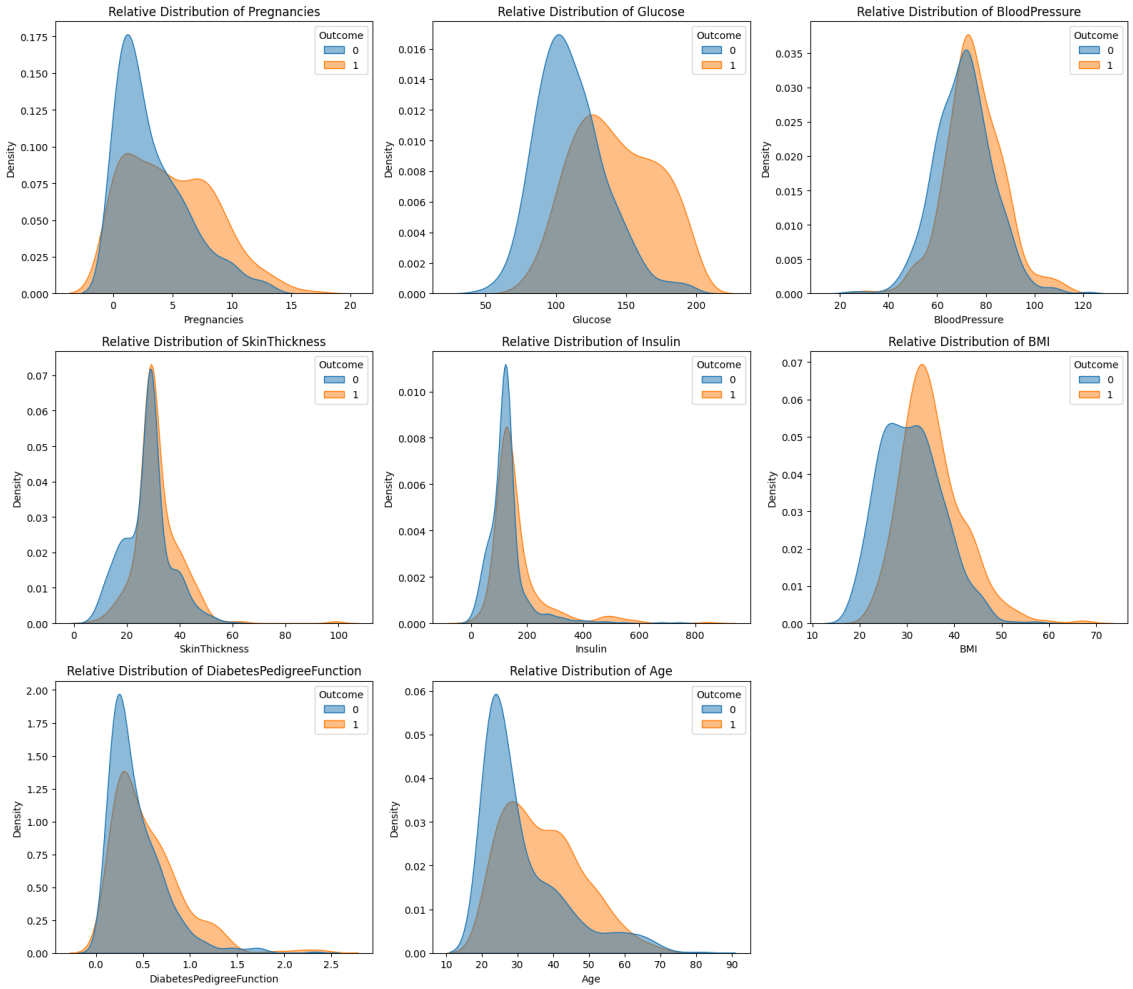


Figure 3: Relative density distribution plot.

- **Age:** Diabetes probability increases with age, older tail dominated by `outcome` = 1. This implies, Age is a progressive metabolic risk.
- **DiabetesPedigreeFunction:** Long right tail for diabetics, indicates genetic susceptibility. This means Pedigree is a non-obvious but important signal.
- **Insulin and SkinThickness:** Heavier overlap, still informative in non-linear models (XGBoost). They both are less predictive alone.
- **BloodPressure:** Significant overlap, weak standalone predictor. It is a secondary risk factor.
- **Pregnancies:** Higher pregnancies → higher diabetes risk, reflects gestational diabetes history. It is useful for population-specific insight.

6 XGBOOST + SHAP

We now change the model to XGBoost for a multitude of reasons including its ability to capture nonlinear effects, handle features automatically, produce strong recall score, and integrate perfectly with SHAP. Results obtained using the `XGBoost` model are depicted in Figure 4:

```
[[78 22]
 [14 40]]
              precision    recall  f1-score   support

           0       0.85      0.78      0.81       100
           1       0.65      0.74      0.69        54

    accuracy                           0.77       154
   macro avg       0.75      0.76      0.75       154
weighted avg       0.78      0.77      0.77       154

ROC-AUC: 0.830925925925926
```

Figure 4: Calculations with `XGBoost` classifier for PIDD.

Figure 4 tells us that `XGBoost` very strong for early detection because:

- Recall for the diabetes class has increased — good sign
- ROC-AUC has increased — good sign
- Fewer false negatives than LR and RF

Next, we combine `XGBoost` with `SHAP` as the merger is claimed to be the **gold** standard for explainable ML on tabular data.

7 GLOBAL FEATURE IMPORTANCE (SHAP)

Note that the train-test split was done as before and we used the training data to do summary plot of the SHAP values, see Figure 5.

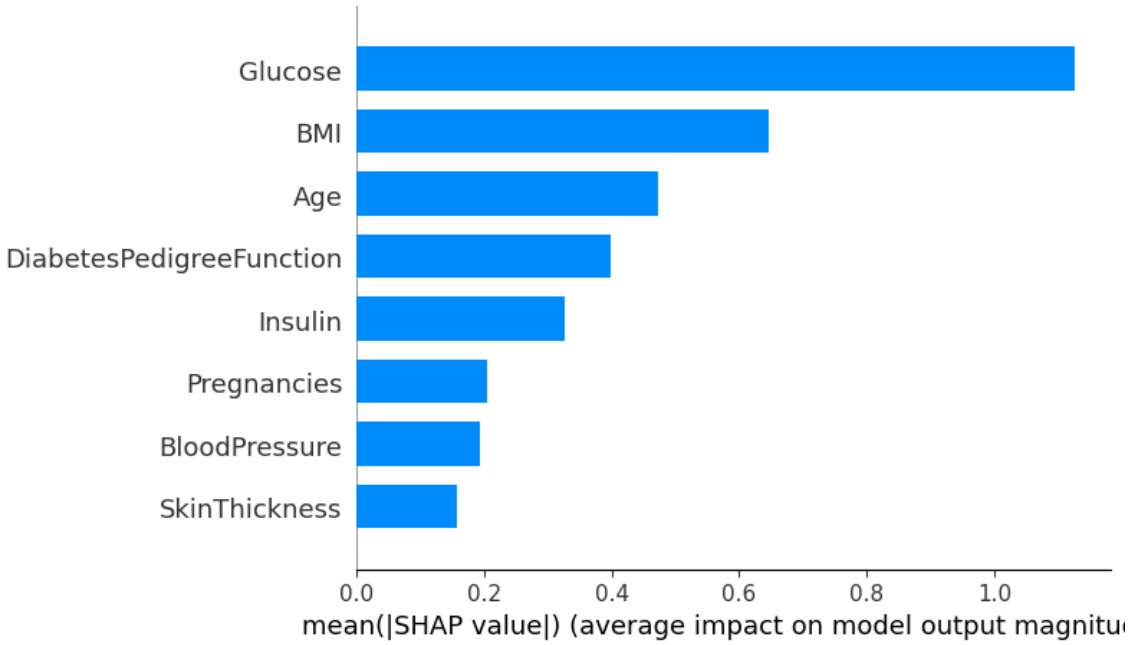


Figure 5: Summary of SHAP values obtained from `XGBoostClassifier()` + SHAP for PIDD.

The results depicted in Figure 5 tell the following ranking for the risk factors:

1. Glucose (dominant early biomarker)
2. BMI
3. Age
4. DiabetesPedigreeFunction
5. Insulin

This confirms medical plausibility, which is critical in healthcare ML.

8 DIRECTION of RISK ('BEESWARM' PLOT)

A nice looking 'bee swarm' plot is depicted in Figure 6 which is interpreted as:

- High glucose $-->$ strong push toward diabetes
- High BMI + older age $-->$ nonlinear risk amplification
- Low values $-->$ protective effect

This shows how risk changes, not just which features matter.

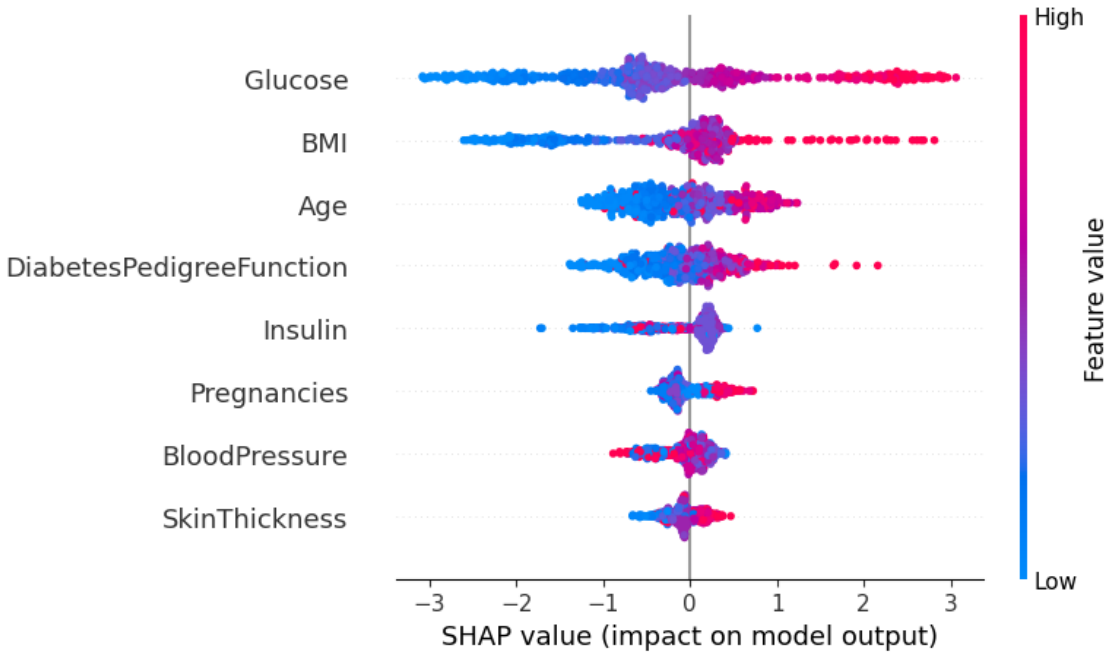


Figure 6: 'Beeswarm' plot of the summarized SHAP values using `XGBoostClassifier()` for PIDD.

9 RISK-THRESHOLD VISUALIZATIONS

Now we visualize how the risk score translates into decisions. We will use the `XGBoost` risk scores.

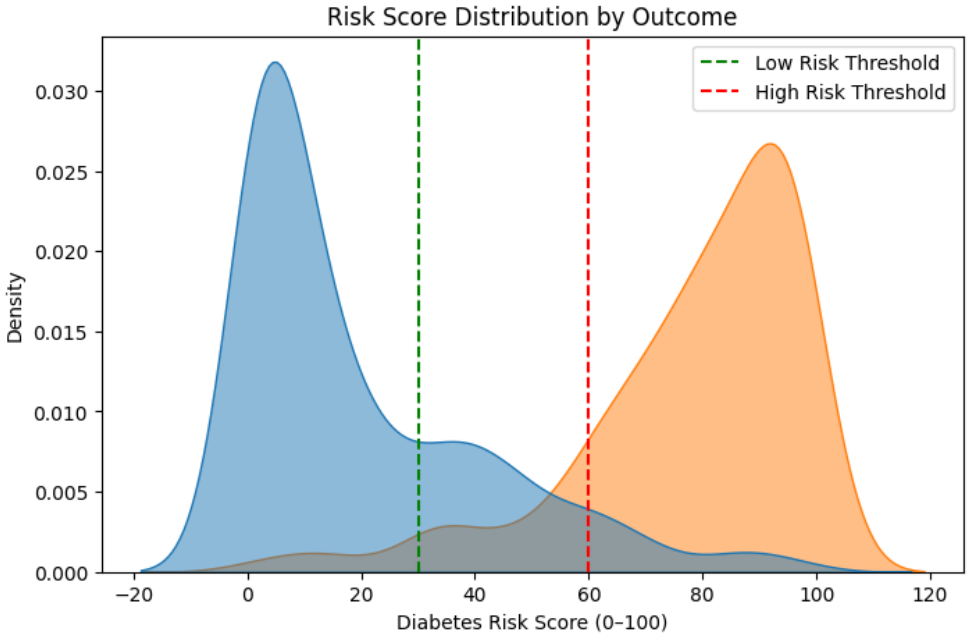


Figure 7: Risk-threshold visualizations for screening decisions.

The results in Figure 7 are interpreted as:

- Diabetics dominate right side
- Non-diabetics dominate left side
- Overlap region implies diagnostic uncertainty
- Thresholds convert probabilities into actionable rules

10 XGBOOST-BASED DIABETES RISK SCORE

The diabetic risk scores have been calculated for each individual and depicted in Figure 8 after arranging them in categories.

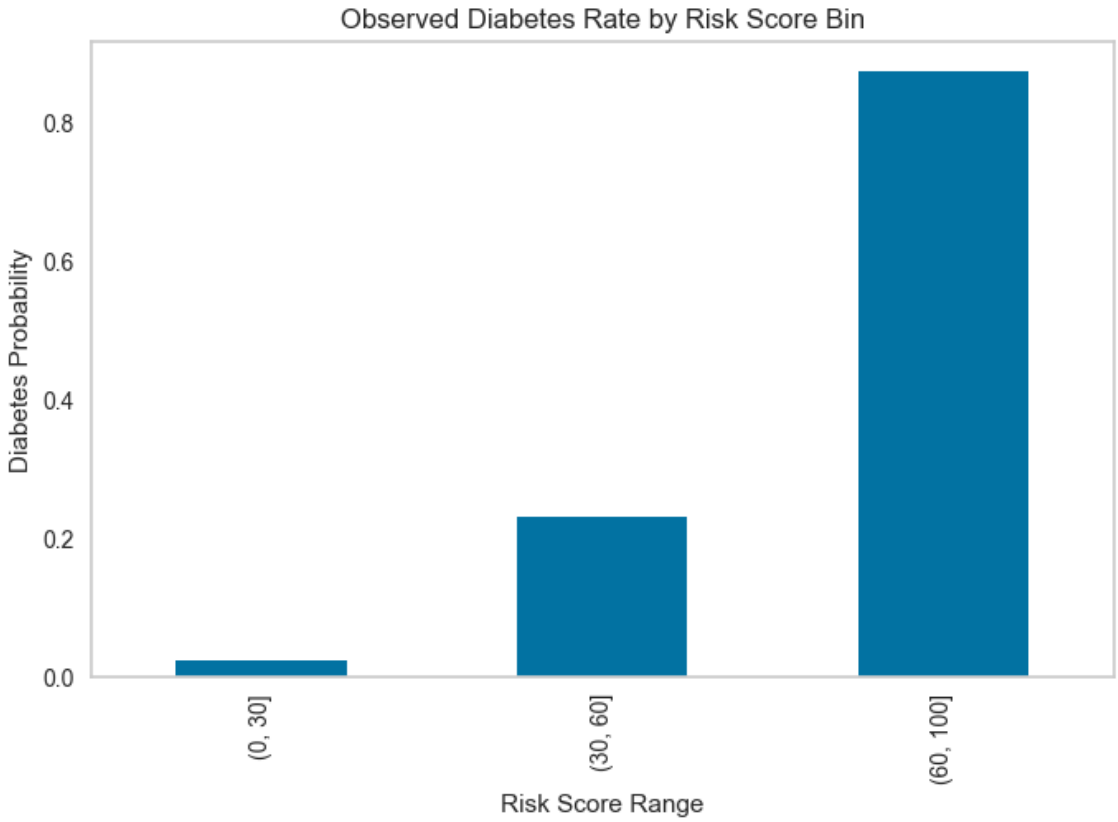


Figure 8: Risk scores vs probabilities of diabetes for PIDD.

Figure 8 tells:

- Diabetes rate increases monotonically
- High-risk bins have very high prevalence
- Confirms good calibration

- Ideal for population screening

The categorization has been done as per the risk score of the patients and summarized in Table 4.

Table 4: Summary of risk score per category.

XGB score range	Risk category	Meaning	Number of patients
< 30	Low risk	Unlikely diabetic	379
$[30, 60)$	Moderate risk	Monitor closely	265
$>= 60$	High risk	High likelihood – recommend screening	124

The risk categorization has also been validated and found out the following:

- **High-risk group** → highest diabetes prevalence
- **Low-risk group** → very low prevalence
- Confirms suitability for **screening programs**

11 NMF PHENOTYPES AND XGBOOST + SHAP FEATURES

For employing Nonnegative Matrix Factorization (NMF) on Pima, the first thing to do is to prepare the data matrix. This starts with selecting the feature columns of the data (all except Outcome) and ensuring non-negativity by applying the `MinMaxScaler()`. Eventually, fit NMF after choosing an appropriate rank k and create proper data frames for the factors W and H.

For choosing the 'optimal' number of clusters k, we first employed the elbow method and obtained $k = 5$ as a starting value, see Figure 9.

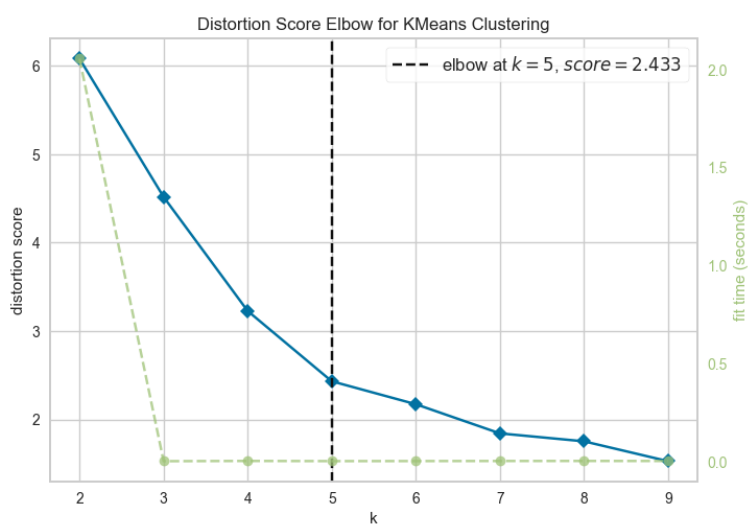


Figure 9: The elbow method.

With the help of the elbow method, five latents (hidden features) were identified and depicted in the above figure and described in Table 5.

Table 5: Formal naming of latent factors.

Latent	Dominant Features	Formal Name
`latent_1`	Glucose, Insulin, BP	Insulin Resistance and Cardiometabolic Stress Phenotype
`latent_2`	Pregnancies, BMI, BP	Reproductive–Obesity Metabolic Load Phenotype
`latent_3`	Pedigree, BMI, Glucose	Genetic Susceptibility–Driven Glycemic Phenotype
`latent_4`	BP, Pregnancies, Skin Thick.	Adipose Distribution and Vascular Reactivity Phenotype
`latent_5`	Age, BP	Age-Related Vascular Risk Phenotype

The practical interpretation of the results presented in the above table is that one needs to do screening as: High `latent_1` — immediate glucose testing, High `latent_3` — family-history-based screening, High `latent_4` — lifestyle intervention.

It is also worth noticing that NMF and XGBOOST + SHAP are complimentary not competing. Table 6 below presents what each method explains.

Table 6: NMF, XGBoost + SHAP and explanations.

Method	**Answers**
NMF	Which biological pathways exist in the population?
XGBoost + SHAP	Which features drive predictions for individuals?

Table 7 below maps NMF phenotypes with SHAP features. The insight we get from this table is that SHAP confirms what NMF discovered structurally but SHAP is based on feature-level while NMF is based on pathway-level.

Table 7: NMF, XGBoost and explanations.

NMF Phenotypes	**Corresponding SHAP Features**
Insulin Resistance & Cardiometabolic	Glucose, Insulin, BP
Reproductive–Obesity	BMI, Pregnancies
Genetic–Glycemic	Pedigree, Glucose
Adipose–Vascular	Skin Thickness, BP
Age-Vascular	Age, BP

12 NMF AND THE BEST SILHOUETTE SCORE

Employing the highest silhouette score strategy, on top of the elbow method, provided $k = 3$ as the best value for the number of clusters, see Figure 10, which contains the numerical values of the latent cluster phenotype profiles. We have clustered patients in latent (phenotype) space, so each cluster

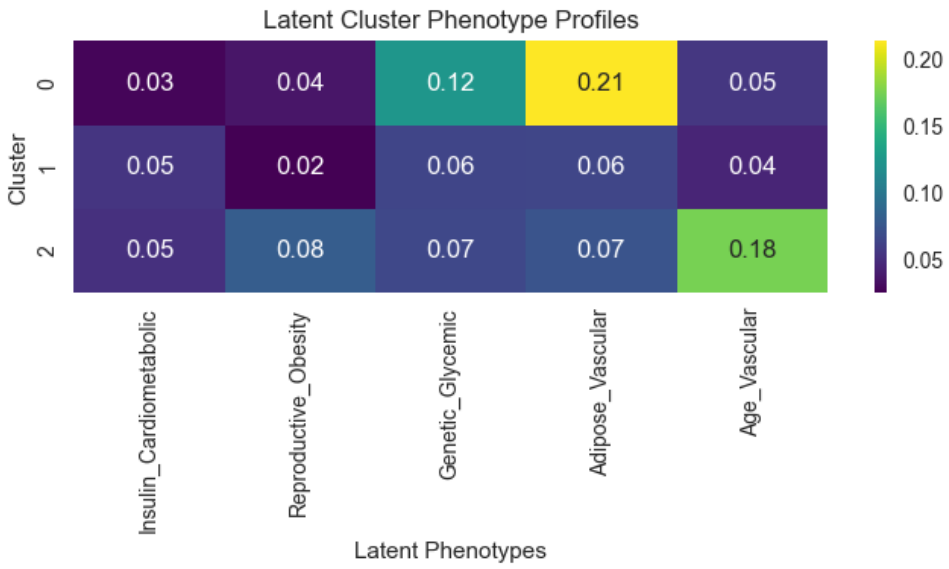


Figure 10: Relative violin plots.

reflects dominant pathway activation, not a single feature. The results are depicted in Figure 10 and interpreted as follows:

Cluster 0 - **Adipose–Vascular Phenotype**: Dominant latent: Adipose_Vascular (Blood Pressure - Skin Thickness - Pregnancies)

Medical meaning: This cluster represents patients with

- Prominent subcutaneous fat distribution
- Vascular reactivity (blood pressure)
- Metabolic stress related to pregnancy history

These individuals are often:

- Pre-diabetic or early-stage
- Not necessarily hyperglycemic yet
- Showing early vascular–adipose dysregulation

Think of this cluster as a "pre-diabetes structural phenotype".

Cluster 1 - **Genetic–Adipose Mixed Phenotype**: Dominant latents: Genetic_Glycemic, Adipose_Vascular. **Medical meaning**: This is a compound-risk group:

- Strong genetic susceptibility (pedigree)
- Elevated glucose
- Amplified by fat distribution and vascular stress

These patients:

- May develop diabetes with less obesity
- Often show earlier onset
- Are high-risk even at moderate BMI

This explains lean diabetics and familial diabetes cases.

Cluster 2 - **Age-Related Vascular Phenotype**: Dominant latent: Age_Vascular.

Medical meaning: This cluster reflects:

- Aging-related insulin sensitivity decline
- Vascular stiffness (blood pressure)
- Diabetes risk driven by age, not obesity

Typical profile:

- Older patients
- Moderate glucose elevations
- Often overlooked in early screening

Think late-onset, vascular-driven diabetes.

Adipose–Vascular in multiple clusters (very important). Why Adipose–Vascular Appears in Multiple Clusters (Very Important)? This is expected and biologically correct. Blood pressure and fat distribution are:

- Shared downstream stress markers
- Not primary disease causes
- Amplifiers of multiple pathways

NMF captures this by:

- Allowing partial membership
- Letting one latent contribute to multiple sub-types

Diabetes is not one disease — shared vascular stress is a unifying theme. The redundancy reflects real metabolic overlap. If clusters were completely disjoint, that would actually be less realistic.

13 CLUSTER SPECIFIC SHAP

In this section, we will explain how XGBOOST behaves inside each NMF phenotype (cluster).

Cluster 0 – Early Adipose–Vascular dysregulation

Insulin_Cardiometabolic_SHAP	0.223
Glucose	0.158
BMI	0.067
Age	0.065
Insulin	0.038

Figure 11: Cluster 0 SHAP summary.

Figure 11 is interpreted as:

- Strongest reliance on insulin–glucose dynamics
- BMI and age play secondary roles
- Indicates early metabolic dysfunction before full obesity or aging effects dominate

Clinical meaning: Diabetes risk is driven by physiological insulin resistance, not lifestyle or aging yet. This is Ideal for early detection and prevention.

Cluster 1 – Genetically Primed Metabolic Diabetes

Insulin_Cardiometabolic_SHAP	0.195
Glucose	0.145
BMI	0.084
Age	0.082
PedigreeFunction	0.050

Figure 12: Cluster 1 SHAP summary.

Figure 12 is interpreted as:

- Glucose and BMI both important
- Genetic signal (Pedigree) is stronger here than other clusters
- Risk arises even at moderate BMI

Clinical meaning: These patients are biologically primed — lifestyle alone may not fully mitigate risk. Supports family-based screening and early pharmacologic prevention.

Cluster 2: – Age-Associated Vascular Diabetes

```
Insulin_Cardiometabolic_SHAP    0.193
Glucose                         0.144
BMI                             0.082
Age                             0.058
PedigreeFunction                0.056
```

Figure 13: Cluster 2 SHAP summary.

Figure 13 is interpreted as:

- Age is more influential than in Cluster 0
- Genetic contribution is also elevated
- Metabolic deterioration accumulates gradually

Clinical meaning: Risk emerges from age-related insulin sensitivity decline rather than obesity alone. Focus on vascular protection and moderate glycemic control.

Why the Patterns Look Similar (Important!) You might wonder: "Why do clusters look somewhat alike?" This is expected and correct because

- All clusters involve Type 2 diabetes
- Glucose and insulin are universal endpoints
- Clusters differ in relative emphasis, not feature exclusion

This indicates that phenotypes differ in why diabetes develops, not what diabetes is.

The big insight. The results show that *XGBoost uses a stable core decision pathway (insulin–glucose), but modulates secondary features differently depending on latent phenotype.* That is a deep, publishable insight.

Conclusion. Cluster-specific `SHAP` analyses demonstrated a shared core decision pathway dominated by insulin–glucose dynamics across all phenotypes. However, secondary feature contributions varied by cluster, with genetic susceptibility contributing more prominently in genetically primed individuals and age exerting greater influence in age-associated phenotypes. These findings indicate that while diabetes prediction relies on a common metabolic axis, risk modulation differs across latent sub-types.

Radar (spider) plots. The radar plots of phenotypes are depicted in Figure 14. Interpretation goes as:

- Each cluster has a distinct shape
- Overlap = shared vascular stress
- Separation = different diabetes mechanisms

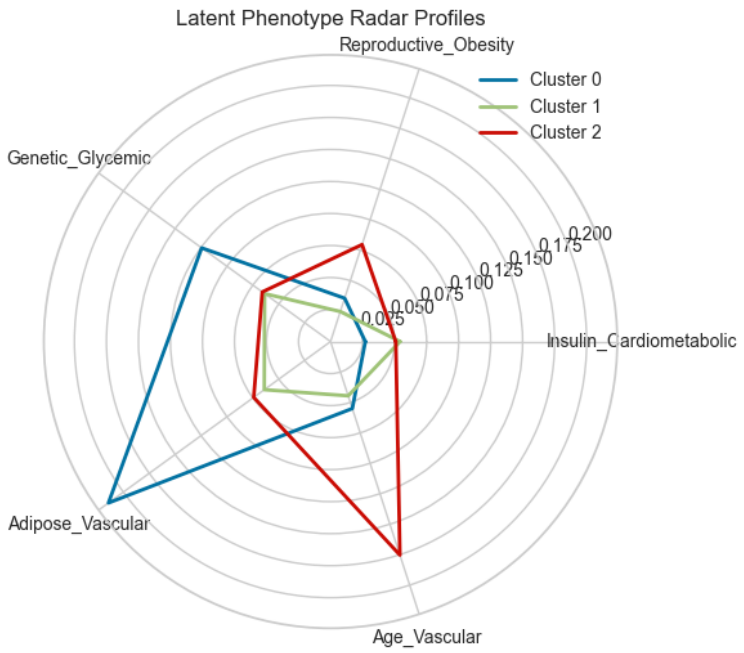


Figure 14: Latent phenotype radar profiles.

14 TARGETED INTERVENTIONS PER CLUSTER

This is where our analysis becomes actionable.

Cluster 0: Early Adipose–Vascular Dysregulation

Primary risks

- Fat distribution
- Blood pressure
- Pregnancy-related metabolic strain

Interventions

- Weight management (moderate, not aggressive)
- Blood pressure control
- Post-pregnancy metabolic monitoring
- Lifestyle intervention —¿ medication

Cluster 1: Genetically Primed Metabolic Diabetes

Primary risks

- Family history
- Hyperglycemia at lower BMI

Interventions

- Early glucose monitoring
- Preventive pharmacotherapy (e.g., metformin)
- Family-based screening
- Lifestyle alone may be insufficient

Cluster 2: Age-Associated Vascular Diabetes

Primary risks

- Aging vasculature
- Blood pressure

Interventions

- Cardiovascular risk management
- Moderate glycemic control
- Avoid overly aggressive weight loss
- Monitor renal and vascular function

Summary Latent phenotype clustering revealed three clinically interpretable diabetes sub-types, each associated with distinct risk profiles and predictive feature importance patterns. Cluster-specific risk scores and SHAP analyses demonstrated that supervised model decision logic varied meaningfully across phenotypes, supporting the utility of phenotype-aware risk stratification and targeted intervention strategies.

15 COMPARISON: NMF VS PCA

In general, PCA is about variance compression while NMF is about latent risk factor discovery. Our experiments reveal that PCA optimizes variance whereas NMF optimizes additive reconstruction. These two are not directly comparable numerically, only conceptually.

Experiments done regarding component loadings reveal the following:

1. PCA loadings

	PC1	PC2	PC3
Pregnancies	0.128432	0.593786	0.013087
Glucose	0.393083	0.174029	-0.467923
BloodPressure	0.360003	0.183892	0.535494
SkinThickness	0.439824	-0.331965	0.237674
Insulin	0.435026	-0.250781	-0.336709
BMI	0.451941	-0.100960	0.361865
DiabetesPedigreeFunction	0.270611	-0.122069	-0.433189
Age	0.198027	0.620589	-0.075248

Figure 15: PCA loadings, numerical values.

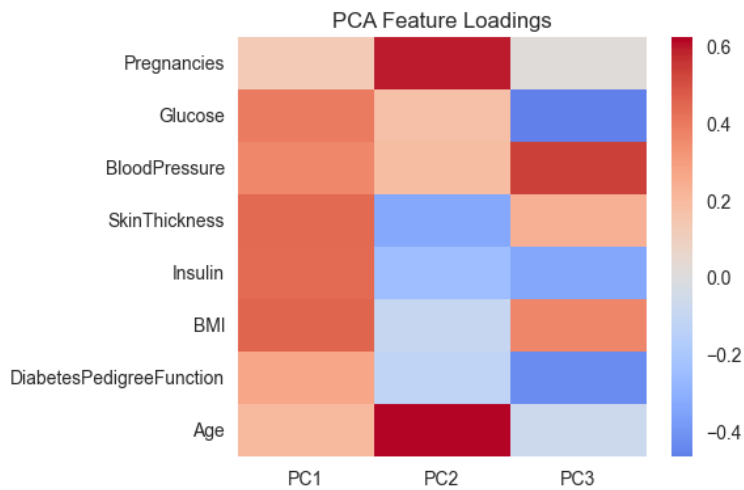


Figure 16: PCA loadings, heatmap.

From Figures 15 and 16 we observe:

- Mixed signs (±)
- Features canceling each other
- No clean phenotype

2. NMF loadings (Clinically Interpretable)

	NMF1	NMF2	NMF3
Pregnancies	0.000000	2.667560	0.138831
Glucose	4.437218	0.206331	1.200371
BloodPressure	3.839590	0.322910	1.258490
SkinThickness	0.000000	0.000000	1.579464
Insulin	0.000000	0.000000	0.786645
BMI	3.023123	0.000000	1.341256
DiabetesPedigreeFunction	0.622925	0.070370	0.769628
Age	0.241011	2.390640	0.000000

Figure 17: NMF loadings, numeric values.

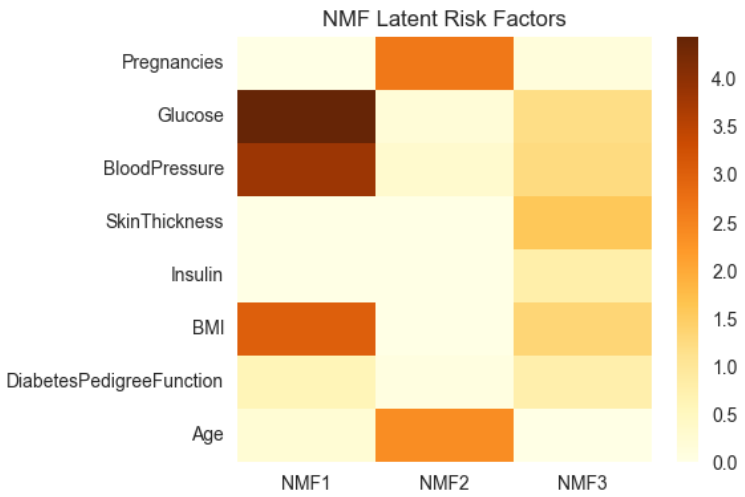


Figure 18: NMF loadings, heatmap.

From Figures 17 and 18 we observe:

- Only positive contributions
- Clear dominance patterns like:
  - i) Glucose + Insulin
  - ii) BMI + SkinThickness
  - iii) Age + Pregnancies

What do we get? *While PCA captured a larger fraction of total variance, its components mixed positively and negatively contributing clinical features, limiting interpretability. In contrast, NMF produced non-negative, additive latent factors corresponding to clinically coherent metabolic phenotypes. Despite similar predictive performance, NMF offered superior interpretability and phenotype stability, making it more suitable for risk stratification and intervention design.*

Final verdict on the comparison. Table 8 contains the final verdict of the comparison.

Table 8: Final verdict: NMF vs PCA.

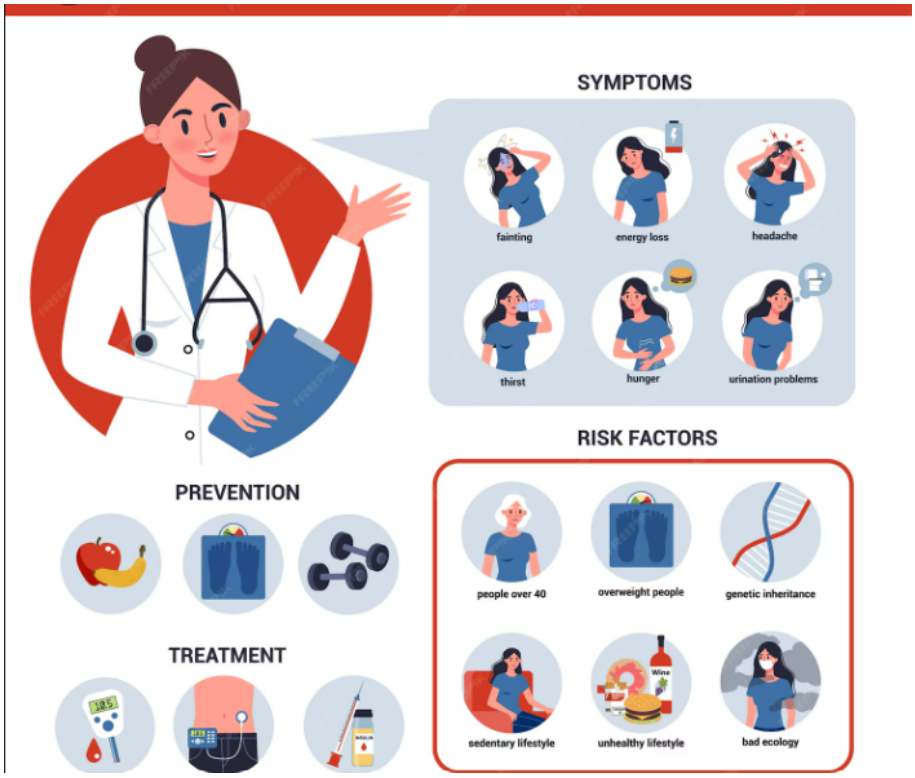
Question	Winner
Variance compression	PCA
Clinical interpretability	NMF
Phenotyping	NMF
`SHAP` alignment	NMF
Intervention design	NMF

Recommendation. The final recommendation is that use NMF for:

- Phenotypes
- Risk scores
- Cluster `SHAP`

However, mention PCA as baseline, not main method.

16 INSIGHT: SCREENING POLICY and INTERVENTION



We have found the following:

1. XGBoost is found to be the winner (compared to LR and RF) because of the following reasons:
   - it provides the best recall score — very useful for early detection
   - it makes very good use of the non-linearity effects, i.e., combined effects of the features
   - it facilitates high quality SHAP explanations
   - its results are clinically trust worthy
2. Glucose is the dominant early biomarker, it pushes strongly towards diabetes
3. Combined (non-linear) risk amplification is revealed by older Age + high BMI

4. The recommended screening policy is presented in Table 9.

Table 9: Summary of risk score per category.

Risk score range	Action
< 30	No action
$[30, 60]$	Monitor/lifestyle intervention
> 60	Immediate clinical testing

This policy is:

- Data-driven
- Clinically interpretable
- Early-detection optimized

5. NMF in combination with the elbow method revealed five ($k = 5$) latent metabolic phenotypes underlying diabetes risk, highlighting the heterogeneous pathways through which Type 2 diabetes develops in the Pima Indian population.

6. BP appears in many NMF clusters indicating that it is a systemic stress indicator, it is linked to insulin resistance, obesity, age and pregnancy.

7. BP is a shared downstream effect of metabolic dysregulation. NMF correctly uses it as a cross-cutting amplifier, not a driver.

8. NMF based cluster-specific analyses demonstrated a shared core decision pathway dominated by insulin-glucose dynamics across all phenotypes.

9. NMF enabled to learn that secondary feature contributions varied by cluster, with genetic susceptibility contributing more prominently in genetically primed individuals and age exerting greater influence in age-associated phenotypes.

10. NMF and silhouette score based latent phenotype clustering revealed three ($k = 3$) clinically interpretable diabetes sub-types, each associated with distinct risk profiles and predictive feature importance patterns.

11. Experimenting on diabetes prevalence per cluster revealed that the Genetic-Adipose cluster shows highest diabetes rate while Adipose-Vascular and Age-Associated-Vascular clusters exhibit moderate and moderate to high (depending on age) diabetic behavior, respectively.

12. Cluster-specific risk scores and SHAP analyses demonstrated that supervised model decision logic varied meaningfully across phenotypes, supporting the utility of phenotype-aware risk stratification and targeted intervention strategies.

13. It is recommended to use NMF for phenotypes, risk scores and clustering but mention PCA only as a baseline, not main method.